

The mahjong-tiles Package

Typesetting Riichi Mahjong Hands and Discard Rivers

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Version 2.2.1, 2026-07-02

Abstract

`mahjong-tiles` typesets Japanese/Riichi Mahjong tiles from compact MPSZ notation. It provides a document command, `\mahjong`, for hands and melds, and `\mahjongriver` for discard rivers. It supports red fives, face-down and blank tiles, rotated and stacked tiles, concealed-kong shorthand, local and global size options, and recolouring of tile backs.

1 Loading the package

A minimal document looks like this.

If you are working from the CTAN source archive, generate the runtime package file first:

```
latex mahjong-tiles.ins
```

The distributed user manual lives in `doc/mahjong-tiles-doc.pdf`;

Its source is `doc/mahjong-tiles-doc.tex`;

The package implementation source is `mahjong-tiles.dtx`;

And `mahjong-tiles.ins` extracts `mahjong-tiles.sty`.

```
\documentclass{article}
\usepackage{mahjong-tiles}
\begin{document}
\mahjong{111m456s111p11122z}
\end{document}
```

The package expects the PDF artwork in the `tiles/` directory. With the default setting, `mahjong-tiles.sty` and `tiles/` should be installed in the same TeX-searchable package directory.

2 Package options

Options can be supplied when loading the package or changed later with `\mahjongtilessetup`.

```
\usepackage[
  height=1.5\baselineskip,
  scale=0.75,
  color=blue!70!black,
  aka=1
]{mahjong-tiles}
```

```
\mahjongtilessetup{height=1.5\baselineskip,scale=0.75,color=teal!65!black,aka=1}
```

Option	Default	Description
height	\baselineskip	Height of one upright tile.
scale	0.75	Scale factor for the tile symbol relative to the tile face.
tile-dir	tiles	Directory containing the tile PDF files.
color	none	Recolour the tile back. Use any xcolor colour expression, or none for the embedded artwork.
aka	1	Red-five mode: 0 disables red fives and maps 0m, 0p, and 0s to regular fives; 1 renders them normally; 2 also makes 55555p show an extra red five pin in concealed-kong shorthand.
river-cols	6	Number of tiles per discard-river row.
river-row-gap	0pt	Vertical gap between discard-river rows.

3 The \mahjong command

```
\mahjong{<tiles>}
\mahjong[<height>][<scale>][<back-colour>][<local-keys>]{<tiles>}
\mahjong[<local-keys>]{<tiles>}
```

The first optional argument also accepts a key-value list. This allows compact local overrides such as:

```
\mahjong[height=1.5\baselineskip,scale=0.75,color=teal!65!black,aka=0]{x 0m0p0s}
```

4 MPSZ notation

A tile is written as one or more digits followed by a suit letter. The suits are **m** for manzu, **p** for pinzu, **s** for souzu, and **z** for honors.

Input	Output
111m456s111p11122z	
1112345678999p	
19m19s19p1234567z	

4.1 Special tokens

Token	Meaning
0m, 0p, 0s	Red fives: 
x	Face-down tile: 
?	Unknown blank tile: 
-	Full visual gap between groups.
N-	Proportional gap of N/7 of one tile width, for example 2-.

4.2 Rotated and stacked tiles

The marker ***** or an apostrophe rotates the preceding tile sideways. The marker **+** or a double quote stacks two sideways copies of the preceding tile, which is useful for kan notation.

Input	Output
111m111s111p22z2-3*333z	
123m-xx?4p'5p"0s	

4.3 Red fives and the aka option

Red fives are controlled by the integer **aka** option. The default **aka=1** renders 0m, 0p, and 0s as red fives. Set **aka=0** to suppress red fives and render those inputs as regular fives. Set **aka=2** to keep normal red fives and add one more red five pin in the 55555p concealed-kong shorthand.

Input	Output
0m0p0s55555m	
\mahjong [aka=0]{0m0p0s55555p}	
\mahjong [aka=2]{55555p}	

4.4 Concealed kongs

Five identical consecutive digits followed by a suit are rendered as a concealed kan: a face-down tile, two visible tiles, and a face-down tile. For suited fives, the visible pair is controlled by **aka**: **aka=0** gives two regular fives, **aka=1** gives a red five plus a regular five, and **aka=2** gives 55555p two red five-pin tiles.

Input	Output
55555p	
33333z	
\mahjong [aka=0]{55555s}	
\mahjong [aka=2]{55555p}	

5 Tile-back recolouring

The tile-back colour can be set globally or locally. The colour value is passed to **xcolor**. The special value **none** restores the embedded back tile.

```
\mahjongtilessetup{color=purple!70!black}
\mahjong{x x x}
```

```
\mahjong[color=teal!65!black]{x x x}
\mahjong[color=none]{x x x}
```



6 Discard rivers

The **\mahjongriver** command uses the same notation as **\mahjong**, but it breaks tiles into rows. The row length is controlled by **river-cols**.

```
\mahjongriver{1m9m2z5z3s6s1s1m8p4z87m}
\mahjongriver[river-cols=6]{1m9m2z5z3s6s1s*1m8p4z87m}
```

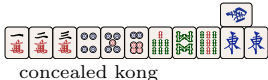


7 Overlay Notes

The `\mahjong` command supports text annotations. Use `[]` in `\mahjong` to attach a note above the previous tile.

Annotations may contain ordinary TeX material. Mahjong tiles inside an annotation are rendered only when they are explicitly wrapped in `\mj`. For example, `[waiting \mj3m]` prints the text "waiting" followed by a small mahjong tile in the annotation. The `\mj` command is parsed only one level deep inside annotations.

The `\mj` command may also be used directly inside `\mahjong`. In that case, its argument must contain exactly one tile. The tile is typeset as a normal-size rotated tile above the hand, which is useful for showing a drawn tile.

Input	Output
<code>111m111s111p2z2-3*333z7-2z[waiting]</code>	
<code>111m113s111p2z2-3*333z7-2z[discard \mj {3s}],\\ waiting \mj {1s2-2z}]</code>	
<code>123m456p789s\mj {2z}11z</code>	
<code>33333z[concealed kong]</code>	

When a note is added to the fifth tile of a concealed kong shorthand, a warning `Overlay text '<overlay text>'` is attached to the final back tile of a concealed kong. will be triggered. Since a concealed kong is rendered as four tiles, the note is moved to the fourth rendered tile, namely the final back tile.

8 Expl3 interface

For package authors who prefer an expl3-style interface, `mahjong-tiles` also exposes:

```
\ExplSyntaxOn
\mahjongtiles_typeset_hand:n {111m456s111p11122z}
\mahjongtilesriver:n          {111m456s111p11122z}
\ExplSyntaxOff
```

9 Attribution and licence

The LaTeX package code is distributed under the MIT Licence. Portions are based on Daniel Schmitz's `mahjong` package, copyright 2021 Daniel Schmitz, also under the MIT Licence.

The tile artwork in `tiles/` is derived from FluffyStuff's `riichi-mahjong-tiles` project. The upstream licence file places that work in the public domain/CC0. Attribution is retained here and in the README.